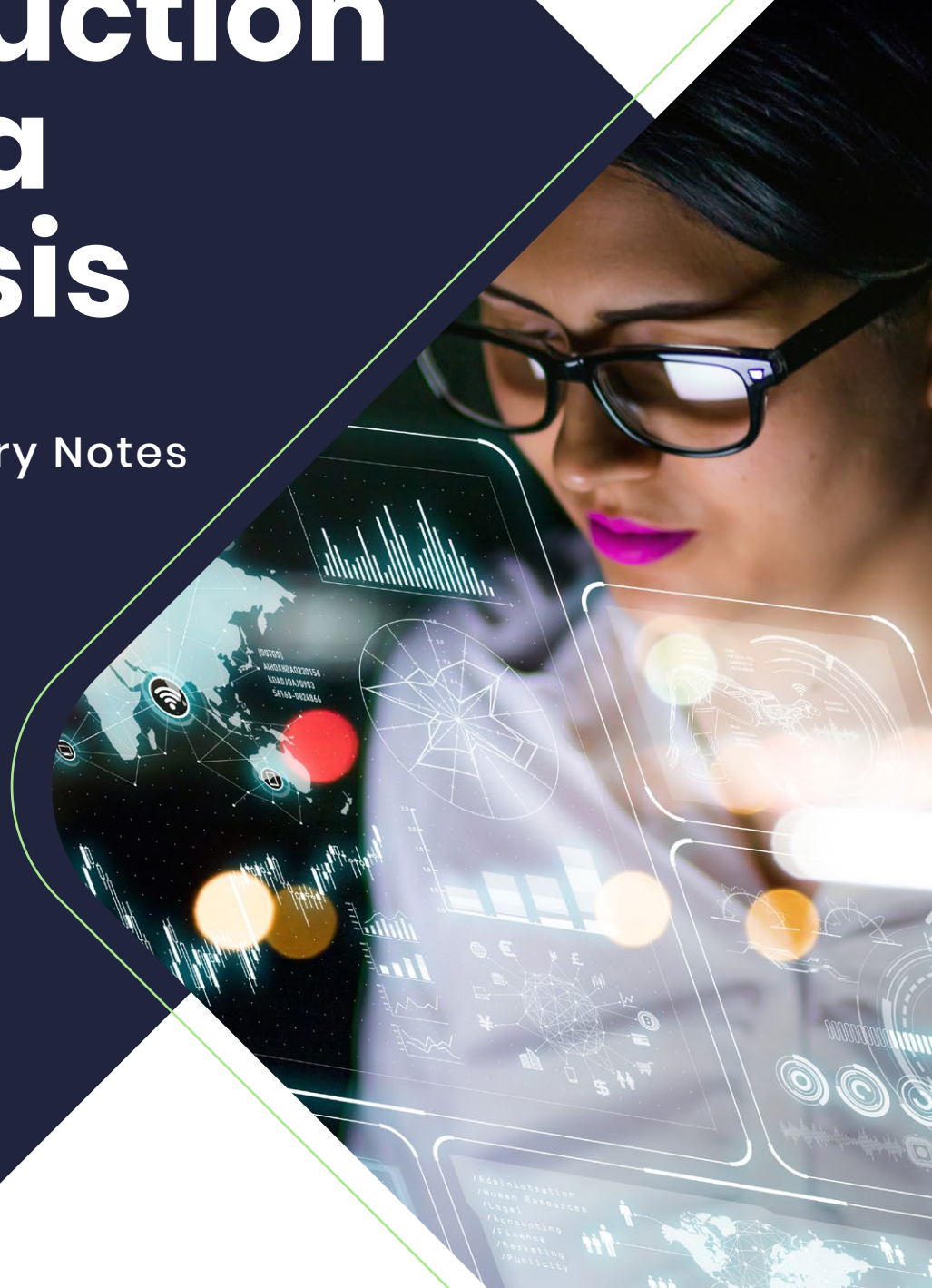


Diploma in Data
Analysis

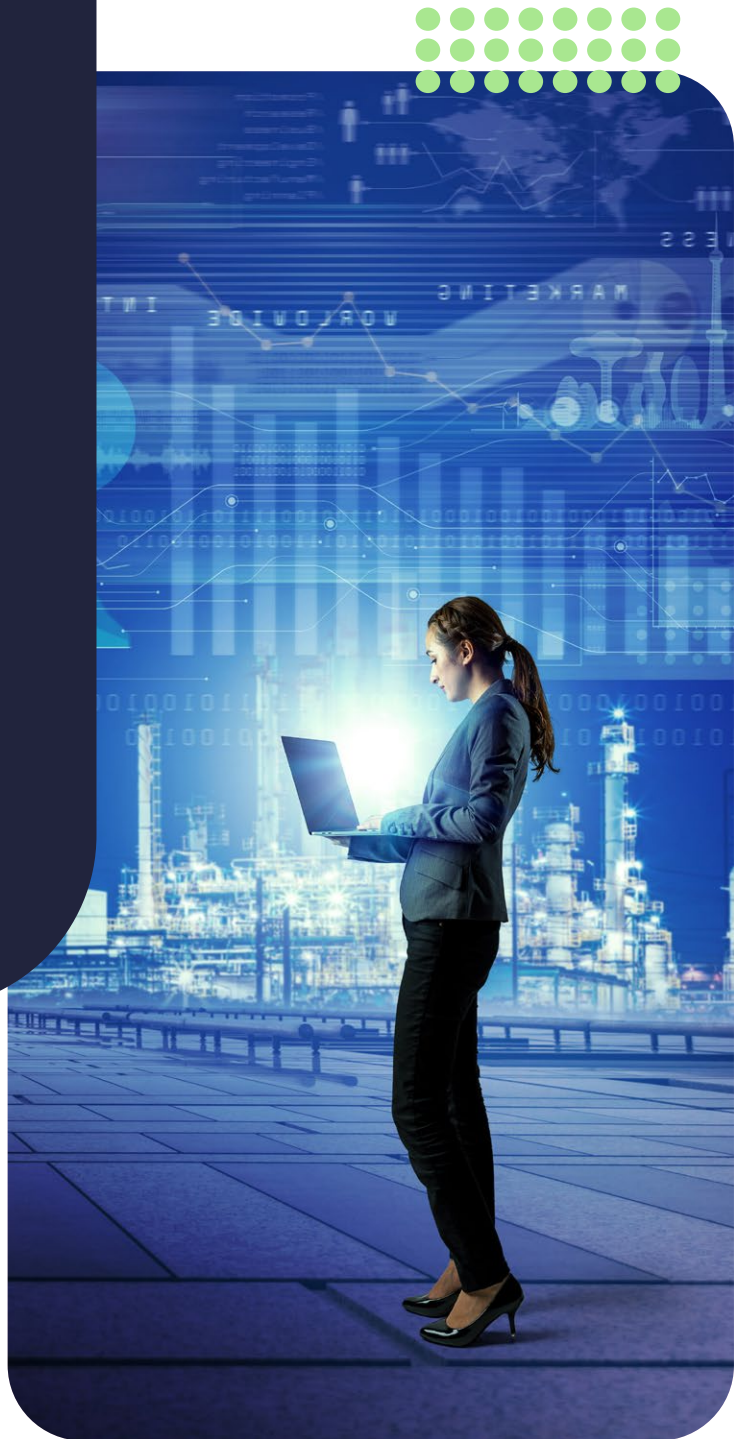
Introduction to Data Analysis

Lesson 3: Summary Notes



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Lesson Objectives

- **Objective 1:** Data analysis Toolpak
- **Objective 2:** Introduction to Probability
- **Objective 3:** Introduction to Random Variables

Lesson Introduction

This part of the journey will introduce you to another tool to add to the Data Analysis Toolkit. You will also add foundational building blocks in the form of probability so that later in your journey, you are able to interpret analysis results with certainty.

It's crucial to have the strongest possible analytics fundamentals to solidify your understanding of analytics and to ensure that you master the art of separating effects from the noise. Probability theory is a building block to understand the probabilistic noise and how to separate it from the effects of the model. Probabilistic thinking is integral to most research fields that requires collection and analysis of data. Lack of knowledge about these concepts, can lead to accidental pitfalls and misleading analysis, therefore it is imperative to gather as much statistical knowledge to avoid these pitfalls.

Continue the data analyst journey and gather as much knowledge as you can throughout this lesson in order to react to information intelligently and avoid overgeneralisation, not mistakenly identify a relationship of correlation for causality, avoid violations of certain statistical assumptions and avoid using the incorrect tools for analysis.



Data Analysis Toolpak

The Data Analysis Toolpak is an add-in feature of Excel. It provides useful data analysis tools for financial, statistical and engineering purposes. The Toolpak is an add in feature of excel, therefore it has to be installed before you can use it. The Toolpak contains preloaded functions that make exploratory analysis in excel much quicker and easier. The Toolpak includes 19 different features like Anova, correlation, covariance, descriptive statistics, exponential smoothing, histograms, etc. Focus on understanding the descriptive statistics and histogram features of the toolpak.

The Data Analysis Toolpak will allow you to utilise your data analysis tools best in order to save time where possible. You want to be as efficient in your analysis process as possible without compromising on the quality of the process. This Data Analysis Toolpak in excel, will help you to carry out more complex statistical analysis on your data much quicker and easier than you are able to create the same complex formulas in the sheet manually.

As a data analyst, it is important to have an understanding of the different measures of spread and central location, but it is not necessary to memorise each formula for them. The data analysis toolpak allows you to perform these actions without knowing the formulas.

Installing the Toolpak

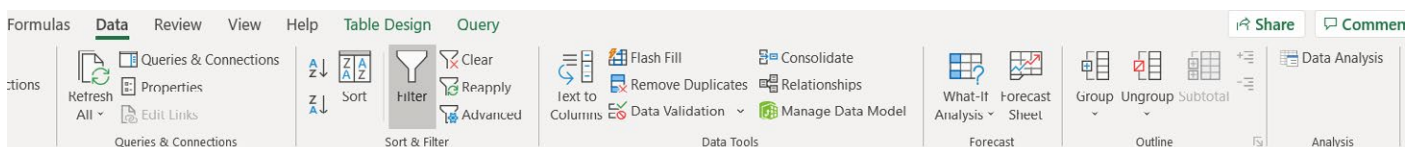
1. Click the File tab, click Options, and then click the Add-Ins category. If you're using Excel 2007, click the Microsoft Office Button , and then click Excel Options
2. In the Manage box, select Excel Add-ins and then click Go. If you're using Excel for Mac, in the file menu go to Tools > Excel Add-ins.
3. In the Add-Ins box, check the Analysis ToolPak check box, and then click OK.
 - If Analysis ToolPak is not listed in the Add-Ins available box, click Browse to locate it.
 - If you are prompted that the Analysis ToolPak is not currently installed on your computer, click Yes to install it.

Just a note

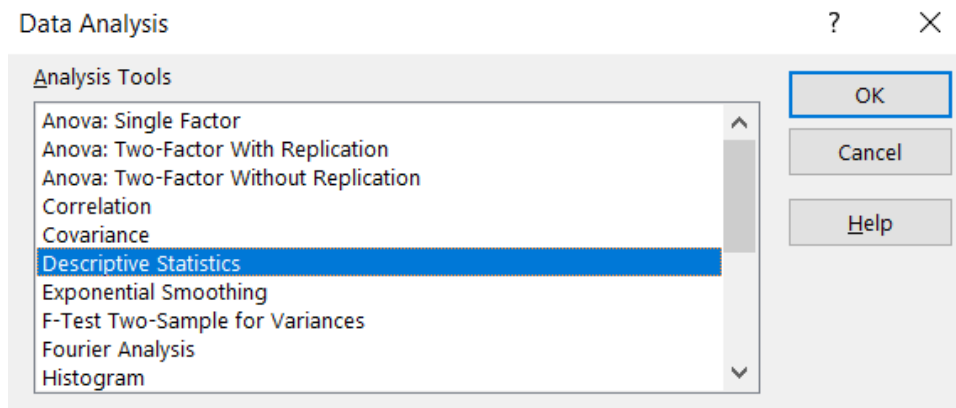
If you are having trouble installing this add-in, please contact: support@shawacademy.com so that our team of experts can help you out and you can practise the examples explored during the lesson yourself.

Investigate the Titanic dataset with the Data Analysis Toolpak

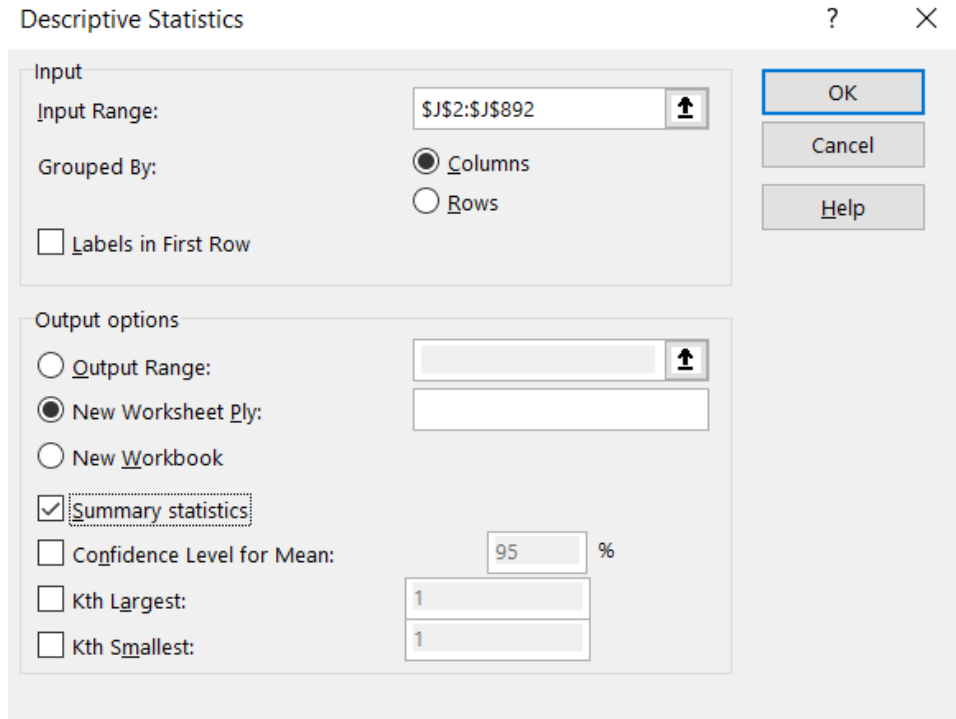
1. Select Data Analysis in Data tab



2. Select Descriptive Statistics



3. Select Input range and Summary statistics



Remember: To change the cell format of the variable to *Number*

NOTES

4. Summary statistics are created on a new worksheet

	A	B
1	<i>Fare</i>	
2		
3	Mean	32,20436588
4	Standard Error	1,664791997
5	Median	14,45
6	Mode	8,05
7	Standard Deviation	49,69341368
8	Sample Variance	2469,435363
9	Kurtosis	33,39831606
10	Skewness	4,787323656
11	Range	512,33
12	Minimum	0
13	Maximum	512,33
14	Sum	28694,09
15	Count	891

- Average of mean passenger fare is approximately £32
- Median passenger fare is approximately £14
- Lowest passenger fare is £0
- Highest passenger fare is approximately £512
- Standard deviation indicates that fare values are spread out over wider range

To note: Minimum value of fare is zero, investigate this further.

Remember:

Standard deviation is a measure of the amount of variation of a set of values. A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range.

5. Create bin range for histogram

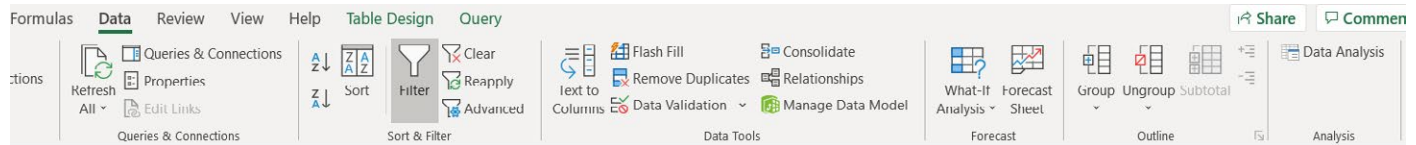
Bins
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150
160
170
180
190
200
210

Bin ranges:

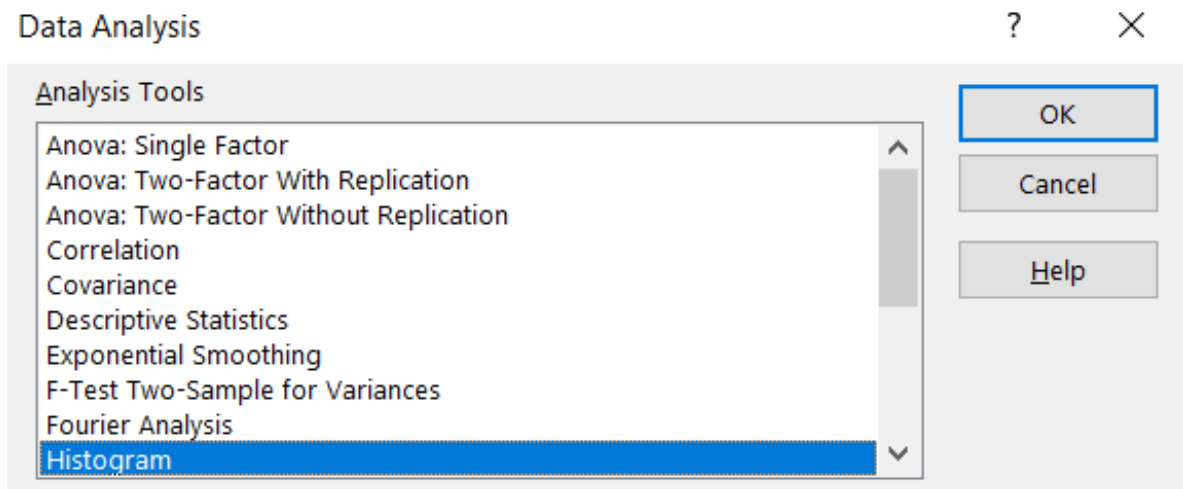
- Bin ranges should be of equal size/width
- Bin ranges should include the minimum and maximum values

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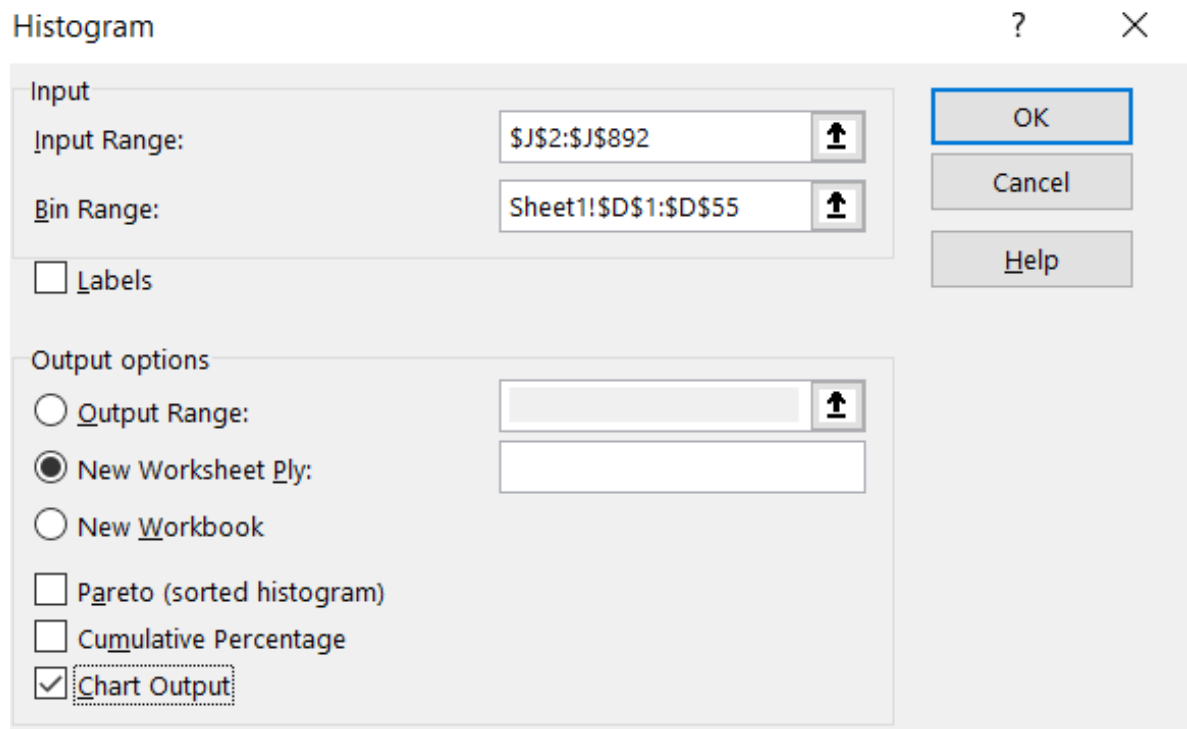
6. Select Data Analysis in Data tab



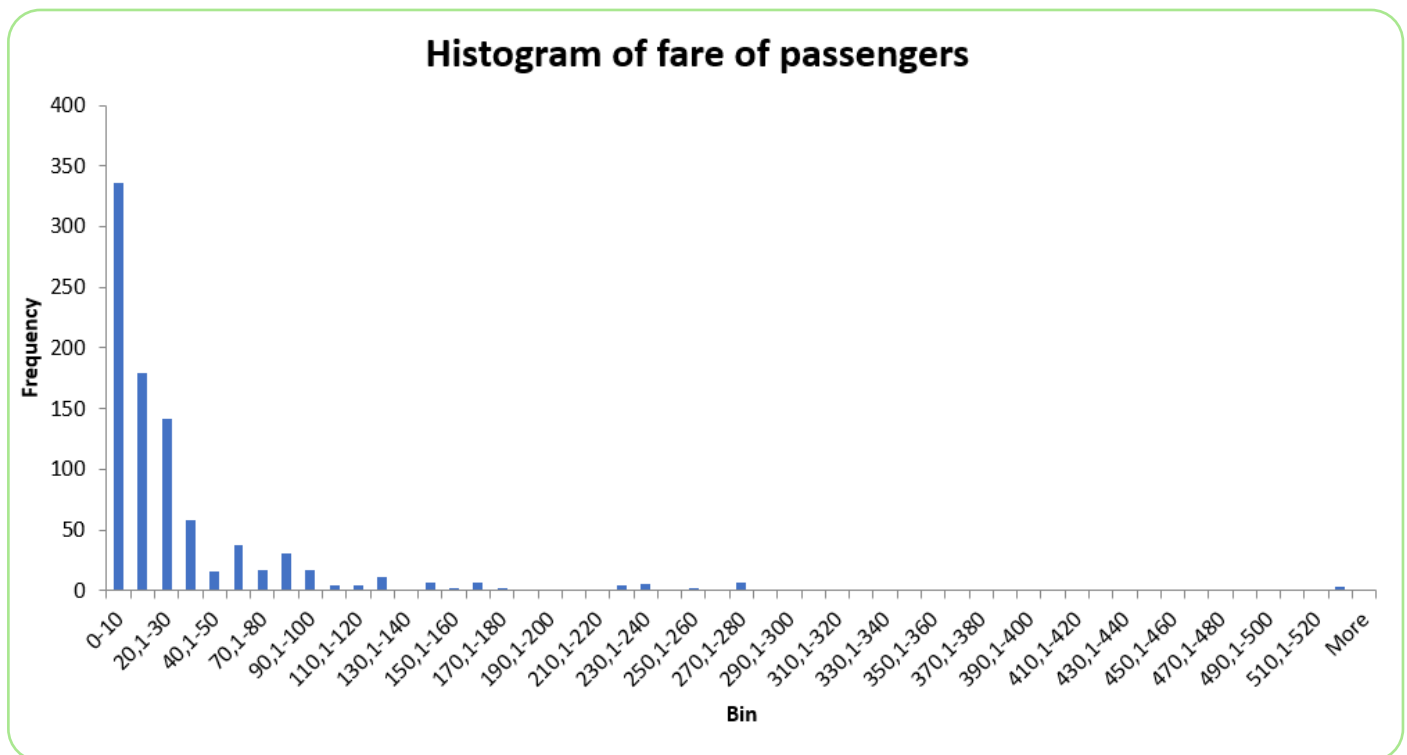
7. Select Histogram option



8. Select input range, bin range and chart output



9. Histogram is creating on another worksheet



- Alter the bins so that it includes a range of numbers.
- Most of the passengers paid between 0 to 30 pounds for their ticket.
- Some values are far removed from the bulk of the data. These observations need to be investigated further and you need to understand why they are so far removed: are they information that is captured incorrectly or is there another reason for the value of these observations?

Pro tip:

Remember, as a data analyst, you do not need to have all the answers, noticing that there might be a problem, is a valuable finding.

Did you know?

Did you know that 100 000 people showed up to see the launch of the titanic?

Probability

Did you know?

Chances of winning the lottery is highly unlikely. If you decide to enter for the powerball draw, there are approximately 42 million different combinations that you can play when choosing 5 out of 50 numbers and 1 out of 20 numbers.

Therefore, if a single play costs you \$1, you need to spend approximately \$42 million for a guaranteed chance to win. The lottery carries what can be referred to as the “house edge”, the only certainty in this game is that the more tickets you purchase, the higher the probability of failure or more likely it is that you will lose.

Probability theory can help you better understand why playing games like the lottery are not in your favour. Probability lays important foundational brickwork for data analysis, and the topic we will be introduced to next.

What is probability?

It's quite an intuitive concept and something you use on a daily basis. You speak and apply probabilities to your everyday life before necessarily realising it. We ask if it will rain almost everyday, if your team will win the toss, if you will win the race. All of these are examples where we incorporate probability to uncertain situations. Probability can be objectively measured, when the number of times an outcome occurs divided by the total number of trials. The outcome that always occurs has a probability of 1 and the outcome that never occurs has a probability of 0.

Probability can also be subjective, one's degree of belief in the occurrence of an event.

Probability can help you better understand the chances of different outcomes occurring. Through this, you can better prepare for the more likely outcome. For example, if you knew there is a high likelihood that it would rain, you will prepare by taking an umbrella or if you knew that you are at a high risk of developing diabetes because of your diet, you will prepare by getting tested and changing your diet.

Terminology

An **experiment** is referred to as the situation that we are uncertain about, that can have more than one outcome. An experiment is also referred to as a trial. It is a procedure that can be repeated and has a set of possible outcomes. The set of possible outcomes can be described as probability.

For example, *if it rains on a weekly basis is an experiment.*

The **outcome** is the possible result of a single trial.

For example, if it rains this week, the outcome of this week's trial from our experiment is that it rained.

An event is one or more outcomes of an experiment.

For example, For our weather example, “it rained” is one of our possible events.

Probability is how likely an event is to occur, it is the measure of how likely an event is. Probability measures the likelihood of an event occurring in a random experiment and is written as a numeric variable between 0 and 1. A number closer to 1 means that the probability of the event is more likely to occur and a number closer to 0 means that the probability is less likely to occur.

Flip the coin

Investigate a simple coin flip example.

The experiment would be the flip or toss of the coin.

If the coin lands on its head, it is one of the possible events for this experiment.

This example is unbiased because the coin is not weighted, therefore the coin has equal chance or probability of landing heads up or tails up, both are equally probable to occur.

It is also not possible for any other outcome to occur, therefore the probability of the coin landing on its head is 0,5 or 50% and the probability of the coin landing on its tail, is also 0,5 or 50%.

The coin is equally (50%) as likely to land on its head as it is as to land on its tail.

Flip the coin 10 times

What happens if the coin is flipped 10 times. Will the probability of landing on either heads or tails still be 0,5?

If it is a fair coin, notice that as the number of coin flips increase, the probability of observing either heads or tails tends to 0,5.

The secret to this phenomenon will be discovered in lessons to come.

Two throws of a dice

Investigate a more complex probability experiment by using 2 dice.

A success is when the sum of the 2 dice numbers facing upwards, is 8.

What is the probability of the sum of the 2 dice being thrown is 8?

How likely is it that the sum of 2 upwards facing dice that are randomly thrown, will be 8?

Sample space

A sample space is a set of all possible outcomes of a random experiment.

The total number of possible outcomes that the rolling of 2 dice have, is 36. Each dice has 1 to 6 numbers of it's faces, and therefore 6 different possibilities, therefore the 2 dice will have 6 squared number of possible outcomes.

The probability of throwing any given total is the number of ways to throw that total divided by the total number of combinations (which is 36 is in this experiment).

(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)
 (2,1) (2,2) (2,3) (2,4) (2,5) (2,6)
 (3,1) (3,2) (3,3) (3,4) (3,5) (3,6)
 (4,1) (4,2) (4,3) (4,4) (4,5) (4,6)
 (5,1) (5,2) (5,3) (5,4) (5,5) (5,6)
 (6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

Throwing the 2 dice at once and sum the value of the upward facing dices, what possible values can the sum of the dice have?

Sum total of 2: (1,1)

Sum total of 3: (1,2); (2,1)

Sum total of 4: (1,3); (3,1); (2,2)

Sum total of 5: (2,3); (3,2); (4,1); (1,4)

Sum total of 6: (3,3); (4,2); (2,4)

Sum total of 7: (3,4); (4,3); (5,2); (2,5); (6,1); (1,6)

Sum total of 8: (4,4); (5,3); (3,5); (6,2); (2,6)

Sum total of 9: (6,3); (3,6); (5,4); (4,5)

Sum total of 10: (5,5); (4,6); (6,4)

Sum total of 11: (5,6); (6,5)

Sum total of 12: (6,6)

Probability of each number

The total number of possibilities remains 36 for rolling 2 dice.

The probability that of an event occurring, is equal to the number of outcomes corresponding to the event divided by the total number of possibilities. Therefore, if the dice lands on 1 and 1, the sumtotal is 2. There is only one outcome for this event (rolling a sum total of 2), therefore the probability of this event is $1/36$.

If the dice lands on 1 and 2 OR inversely 2 and 1, the sumtotal is 3. There are 2 outcomes for this event (rolling a sum total of 3), therefore the probability of this event is $2/36$.



So how many different ways can the sum total of the upward facing dice be equal to 8?

We can throw a (2,6),(6,2),(3,5),(5,3) and (4,4), therefore there are 5 outcomes for this event (rolling a sum total of 8). The probability of this event is $5/36$.

$$2\{(1,1)\} \Rightarrow 1/36$$

$$3\{(1,2),(2,1)\} \Rightarrow 2/36$$

$$4\{(2,2),(3,1),(1,3)\} \Rightarrow 3/36$$

$$5\{(1,4),(4,1),(2,3),(3,2)\} \Rightarrow 4/36$$

$$6\{(3,3),(1,5),(5,1),(2,4),(4,2)\} \Rightarrow 5/36$$

$$7\{(1,6),(6,1),(2,5),(5,2),(3,4),(4,3)\} \Rightarrow 6/36$$

$$8\{(2,6),(6,2),(3,5),(5,3),(4,4)\} \Rightarrow 5/36$$

$$9\{(3,6),(6,3),(5,4),(4,5)\} \Rightarrow 4/36$$

$$10\{(4,6),(6,4),(5,5)\} \Rightarrow 3/36$$

$$11\{(5,6),(6,5)\} \Rightarrow 2/36$$

$$12\{(6,6)\} \Rightarrow 1/36$$

Chances of success for sum of 8?

The chance of success for rolling a sum total of 8 is $5/36$ or approximately 0,14 or 14%,

Therefore, the probability to not roll a sum total of 8, also known as failure, is $31/36$ or approximately 0,82 or 82%.

The chances of achieving success are fairly low.

Also notice that the total probability is always 1, because there are only the 2 options in this experiment, to throw a sum total of 8 or to not throw a sum total of 8.

Core take-aways

Probability tells you how likely it is that an event will occur and is quantified as decimal numbers between 0 and 1.

Therefore, probability cannot be negative.

The higher the chances of the event occurring, the higher the probability and the closer it will be to 1.

Inversely, the lower the chances of the event occurring, the lower the probability and the closer it will be to 0.

The largest possible probability value is 1 and indicates an event that must occur.

NOTES

Random variables

A random experiment is a physical situation whose outcome cannot be predicted until it is observed. A random experiment can be the tossing of a fair coin and the possible outcomes can be heads or tails.

A random variable is a variable with possible numerical outcomes of a random experiment.

A random variable is not the same as the algebra variable, for instance x , but rather a whole set of values. The random variable can take on any of the values in the set. A random variable is defined in terms of the outcome of a process.

In the coin example, denote the random variable as capital X and let X be the set of values that are heads and tails. X can take on heads or tails depending on the outcome of the trail.

There are two types of random variables

1. **Discrete** random variable is one which may take on only a countable number of distinct values such as 0,1,2,3,4, etc
2. **Continuous** random variable is one which takes an infinite number of possible values. Continuous random variables are usually measurements.

Conclusion

These principles, probability and random variables, are foundational to the practise of data analysis. With them securely locked into your brain and the tool you added to your toolkit, you are well on your way to master data analysis.



NOTES

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